GPU-Based Multi-Scale Lightweight Network for Efficient High-Resolution Remote Sensing Change Detection

Shuyi Li *
Jiangsu Automation Research Institute
Lianyungang, China
*Corresponding author: lishuyi_vip@163.com

2025 6th International Conference on Machine Learning and Computer Application (ICMLCA) | 979-8-3503-9243-2/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICMLCA66850.2025.11336424

***Abstract*—High-resolution remote sensing change detection (RSCD) plays a crucial role in various applications. However, traditional methods often face difficulties with high-resolution data due to atmospheric variations, seasonal changes, and intra-class heterogeneity. While deep learning techniques have shown promise, they still struggle with either computational inefficiency or limitations in capturing complex spatial features in large-scale, high-resolution imagery. In this paper, we propose a novel GPU-based method that leverages a Multi-Scale Context Aggregation (MSCA) module to address these challenges. Our approach integrates parallel convolutions with varying dilation rates, image-level pooling, and differential feature processing to expand the receptive field while retaining fine spatial details. This enables our method to effectively model both local changes and long-range contextual relationships without the heavy computational cost of Transformer-based models.**

Experimental results on the Google dataset demonstrate that our method outperforms existing SOTA models, achieving the highest IoU (88.32%), F1-score (93.75%), and Precision (93.45%). Visual analyses further confirm the superiority of our method, showcasing its ability to detect changes with minimal false positives. Loss curve analysis also shows steady convergence, highlighting the stability and efficiency of our model. Our GPU-based MSCA module offers a significant advantage in handling high-resolution RSCD tasks, providing both high accuracy and computational efficiency.

***Keywords*—Remote Sensing (RS), Change Detection (CD), GPU-based.**

I. Introduction

High-resolution remote sensing imagery provides abundant spatial details [1], [2]. Remote sensing change detection (RSCD), which identifies differences across multi-temporal images, is fundamental to applications.

Traditional approaches, including pixel-level differencing [3] and object-based analysis [4], often struggle with high-resolution data where atmospheric variations, seasonal changes, and intra-class heterogeneity produce pseudo detections (Figure 1) and unstable performance [8].

Recent advances in deep learning have substantially improved RSCD [5]. Convolutional neural networks (CNNs) leverage hierarchical representations and multi-scale fusion to achieve robust results. Architectures such as MSCANet [6], and ELGCNet [7] demonstrate the effectiveness of CNNs in extracting discriminative local features [10].

However, conventional convolutions operate isotropically, which limits their ability to capture elongated or irregular structures commonly present in remote sensing imagery. Dilated convolutions extend receptive fields but may introduce grid artifacts, while deformable or directional convolutions improve geometric adaptability yet remain insufficient for comprehensive multi-scale context modeling. At the same time, although Transformer-based methods enhance global dependency modeling, their quadratic time and memory complexity $\mathcal{O}(L^2)$ makes them computationally prohibitive for large-scale high-resolution inputs, hindering practical deployment.

We propose the Multi-Scale Context Aggregation (MSCA) module, specifically designed for Remote Sensing Change Detection (RSCD). Unlike previous techniques, such as dilated convolutions and traditional spatial convolutions, the MSCA module employs a novel architecture that leverages multiple convolutional kernels of varying sizes to perform parallel operations. This allows the model to capture contextual information at multiple scales simultaneously. Additionally, the MSCA module incorporates image pooling operations to aggregate global information, enhancing the model's ability to capture large-scale patterns and features. More specifically, the MSCA module combines parallel convolutions with different dilation rates, enabling the network to extract both local and global features effectively. The integration of image-level pooling further strengthens this capability by ensuring that global context is preserved and merged with fine-grained local details. The combination of differential feature processing and multi-scale convolution operations helps expand the receptive field, facilitating the detection of large-scale changes, while still retaining crucial spatial details. This balanced approach ensures that the model can effectively capture both the fine-grained spatial information and the broader contextual information necessary for accurate change detection in remote sensing imagery.

This design enables simultaneous modeling of local changes and long-range contextual cues without incurring the heavy computational burden of Transformer-based architectures. In

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:45:10 UTC from IEEE Xplore. Restrictions apply.

this way, the proposed method achieves accurate and efficient change detection.

The subsequent comprehensive comparative experiments further validate the performance and effectiveness of the proposed MSCA module. These experiments demonstrate that the MSCA module not only outperforms traditional methods but also provides significant improvements in accuracy and robustness. Additionally, the visualizations of the results, along with the observed trend of decreasing loss, confirm the convergence behavior of the module during training. The loss curve indicates stable learning and consistent reduction in error over time. Furthermore, the MSCA module's ability to effectively reduce pseudo-change detection is highlighted, showcasing its potential in handling noisy or irrelevant data. This combination of high performance, stable convergence, and strong resistance to false positives underlines the promising application prospects of the MSCA module in real-world remote sensing change detection tasks.

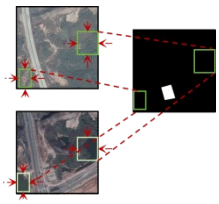


Figure.1 Pseudo-change example diagram.

II. Methodology

In this section, we present the proposed change detection network, which consists of three major components: the overall framework, the Multi-Scale Context Aggregation (MSCA) module, and the multi-scale decoder. The overall framework defines the model's structure and workflow, while the MSCA module aggregates contextual information across multiple scales to enhance feature representation. The multi-scale decoder then refines these features to produce accurate change detection predictions. Each component plays a vital role in improving the network's performance and efficiency.

A. Overall Architecture

Given a pair of bi-temporal remote sensing images $I^A, I^B \in \mathbb{R}^{H\times W\times 3}$, we first employ a backbone network (ResNet-18) to extract hierarchical features at multiple levels. For each stage i, the features are denoted as F_i^A and F_i^B for image I^A and I^B, respectively. To capture the temporal discrepancies, we compute the absolute difference between features:

$$D_i = |F_i^A - F_i^B|, \quad i \in \{1,2,3,4\}. \tag{1}$$

These differential feature maps $\{D_1, D_2, D_3, D_4\}$ are then fed into the Atrous Spatial Pyramid Pooling (MSCA) modules to enrich multi-scale contextual representations. Finally, the decoder aggregates features from different levels and predicts the final change map $M \in \mathbb{R}^{H\times W}$.

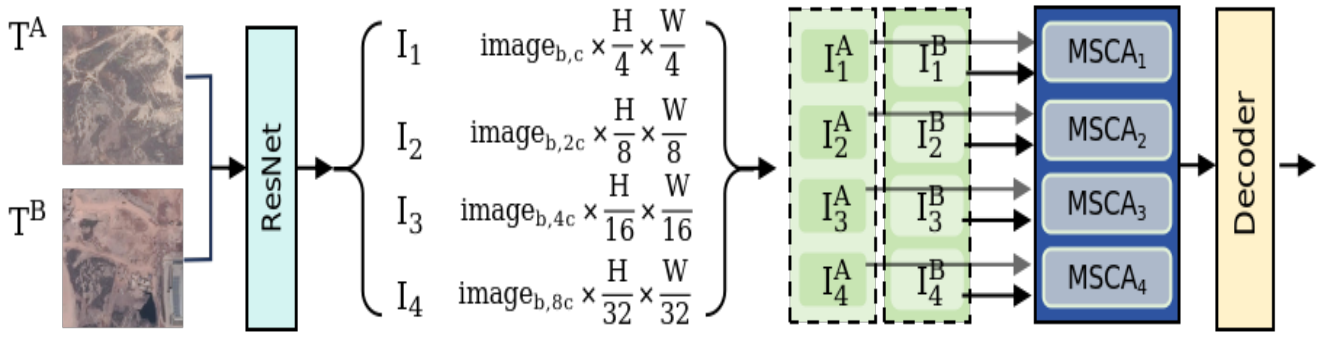


Figure.2 The overall pipeline of the proposed method.

B. Multi-Scale Context Aggregation (MSCA)

To effectively capture contextual dependencies in remote sensing images, we employ an improved Multi-Scale Context Aggregation (MSCA) module (Figure 2). The input to the module is the differential feature map $D_i \in \mathbb{R}^{C\times H\times W}$. The module consists of three types of parallel branches that encode multi-scale context.

First, a 1×1 convolution branch is applied to compress the channel dimension while preserving local structures:

$$B_1(D_i) = \sigma\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times 1}(D_i)\right)\right), \tag{2}$$

where $\mathrm{BN}(\cdot)$ denotes batch normalization and $\sigma(\cdot)$ is the ReLU activation.

Second, an image-level pooling branch is introduced to aggregate global context across the entire image. This is achieved by performing adaptive average pooling, which allows the model to capture the overall spatial characteristics of the input image regardless of its size. The pooled feature map is then passed through a 1×1 convolutional layer, which helps to reduce the dimensionality and refine the feature representation. Finally, the output is upsampled to match the original image resolution, ensuring that the global context is appropriately integrated with the local features for precise prediction:

$$B_2(D_i) = \mathrm{Upsample}\left(\sigma\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times 1}(\mathrm{Pool}(D_i))\right)\right)\right). \tag{3}$$

Third, a set of atrous convolutions with dilation rates $r \in \{1,6,12,18\}$ are employed to enlarge the receptive field and capture multi-scale semantics:

$$B_3^{(r)}(D_i) = \sigma\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times 3}^{(r)}(D_i)\right)\right). \tag{4}$$

The outputs of all branches are concatenated:

$$\mathrm{P_i} = \sigma\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times 1}\left([\mathrm{B_1(D_i)}, \mathrm{B_2(D_i)}, \{\mathrm{B_3^{(r)}(D_i)}\}_{r\in\{1,6,12,18\}}]\right)\right)\right) \tag{5}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:45:10 UTC from IEEE Xplore. Restrictions apply.

In this way, the MSCA module effectively captures both fine local structures and global scene-level information by integrating multi-scale contextual dependencies. The ability to process information at multiple scales allows the model to detect intricate, localized changes, while also considering broader contextual factors that influence the entire scene. This joint approach is particularly crucial in complex remote sensing scenarios, where subtle changes—often small in scale—must be accurately distinguished from larger, more obvious variations. By combining these capabilities, the MSCA module enhances the model's robustness and sensitivity, ensuring it can handle the diverse and dynamic nature of remote sensing data.

C. Multi-scale Decoder

The decoder receives the set of enhanced features $\{P_1, P_2, P_3, P_4\}$ from the MSCA modules. To align feature dimensions, each P_i is first compressed in the channel space. Then, a progressive upsampling and concatenation strategy is applied: higher-level semantic features are upsampled to match the spatial resolution of lower-level features and concatenated with them. This step reduces the number of channels while retaining the essential information for further processing. Next, a progressive upsampling and concatenation strategy is employed. In this strategy, higher-level semantic features, which typically have a lower spatial resolution, are upsampled to match the resolution of the lower-level features with finer spatial details. Once upsampled, these higher-level features are concatenated with their corresponding lower-level counterparts, enriching the feature representation by combining global context with fine-grained spatial details. This process is iterated at multiple levels, progressively refining the features, until the resolution of the shallowest feature map is fully restored. By gradually matching the resolutions, this strategy ensures that both high-level semantic information and low-level spatial details are effectively integrated.

Formally, let F_{dec} denote the fused feature representation after multi-scale aggregation. The final prediction is produced by a classification head:

$$M = \mathrm{Softmax}\big(\mathrm{Conv}(F_{dec})\big), \tag{6}$$

where $M \in \mathbb{R}^{B\times 2\times H\times W}$ represents the binary change detection mask, with B eing the batch size, H and W being the spatial dimensions (height and width) of the output mask, and the second dimension (size 2) indicating the two classes: change and no-change. The Softmax activation function ensures that the final output provides class probabilities for each pixel, enabling the model to classify the corresponding regions as either change or no change.

In summary, the decoder receives a set of enhanced feature maps from the MSCA modules. To ensure uniformity across all feature maps, each feature is first compressed along the channel dimension. Then, a progressive upsampling and concatenation strategy is applied, where higher-level semantic features, which have lower spatial resolution, are upsampled to match the resolution of the finer, lower-level features. These upsampled features are then concatenated with their corresponding lower-level counterparts, combining global context with local spatial details. This process is repeated, progressively refining the features until the resolution of the shallowest feature map is restored. The final fused feature representation is passed through a classification head, where a convolution operation followed by a Softmax function generates the final prediction: a binary change detection mask. The output mask represents class probabilities for each pixel, indicating whether the region corresponds to a change or no change, with two classes—change and no-change—being represented in the output.

III. EXPERIMENTS AND RESULTS

TABLE I. QUANTITATIVE COMPARISONS ON GOOGLE DATASET.

Method	Backbone	Google		
		IoU (%)	F1-score (%)	Precision (%)
MSCANet [6]	ResNet-18	**86.38**	**92.69**	92.25
ELGCNet [7]	ViT-based	77.03	87.03	84.53
ECTI [8]	ResNet-18	85.94	92.43	93.10
BIFA [9]	SegformerB0	81.22	89.64	86.38
DAMFANet	ResNet-18	86.11	92.53	92.90
Ours	ResNet-18	**88.32**	**93.75**	**93.45**

A. Experimental Setup

We evaluate our approach using the Google Dataset, which consists of 19 seasonal image pairs captured from suburban Guangzhou between 2006 and 2019. These images have a spatial resolution of 0.55 meters and are composed of three spectral bands. The dataset is preprocessed by cropping the images into patches of size 512 × 512 pixels. The cropped patches are then split into training, validation, and test sets with a ratio of 500:144:72, ensuring a balanced distribution for model evaluation.

The model is trained for a total of 300 epochs with a batch size of 10. The optimization is performed using the Adam optimizer, which is widely recognized for its effectiveness in training deep learning models. The learning rate for optimization is carefully chosen to ensure stable convergence, though the exact value is omitted for brevity. Additionally, the decoder dimension is set to 64, balancing the trade-off between computational efficiency and model complexity. The entire training process is conducted on an NVIDIA RTX 6000 GPU, providing the necessary computational power for efficient model training.

To assess the performance of our approach, we employ a set of evaluation metrics: Intersection over Union (IoU), F1-score, and Precision. These metrics together provide a comprehensive evaluation of the model's performance. IoU quantifies the spatial consistency of the predicted change detection areas, measuring how well the predicted regions overlap with the ground truth. F1-score strikes a balance between false positives and false negatives, providing insight into the overall accuracy of the model in identifying change events. Finally, Precision evaluates the reliability of the positive predictions, ensuring that the model correctly identifies significant changes without excessive false alarms.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:45:10 UTC from IEEE Xplore. Restrictions apply.

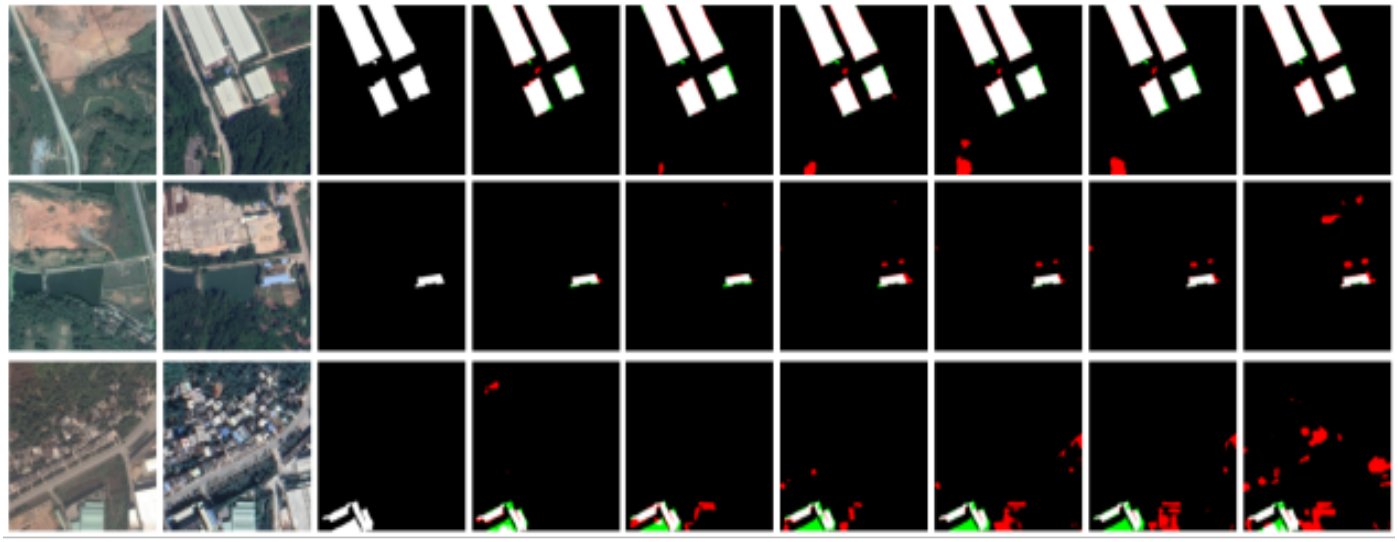


Figure 3. The visualization results on the GoogleBuilding dataset. From left to right, the results of the following methods are displayed: Input A, Input B,Ground Truth, Ours, BIFA, DAMFANet, MSCANet, ECTI, ELGCNet

B. *Performance Comparison*

The quantitative comparison results on the Google dataset are shown in Table 1. Our proposed method achieved the best performance in IoU (88.32%), F1-score (93.75%), and Precision (93.45%), significantly outperforming all other baseline models, especially in F1-score and Precision. Compared to MSCANet and DAMFANet, our method demonstrated notable improvements in both IoU and F1-score, confirming its advantages in the change detection task. Furthermore, although DAMFANet performed well on certain metrics, our method showed a more comprehensive advantage.

Additionally, the visual results, as shown in Figure 3, demonstrate that our model provides a cleaner and more accurate detection of changes, with significantly fewer false positives compared to existing methods. These visual insights are consistent with the improvements observed in the IoU and Precision metrics, which further validate the effectiveness of our approach. The gains in these metrics are visually reflected in the clearer delineation of change regions, indicating that the model is able to distinguish subtle changes with greater precision. Furthermore, the loss curve analysis in Figure 4 reveals a steady and consistent decrease in loss throughout the training process, further confirming the stability of the model's convergence. This behavior suggests that our model not only achieves robust quantitative performance but also exhibits strong generalization capabilities, making it highly effective for real-world change detection tasks. The combination of improved detection accuracy, stable training dynamics, and practical performance indicates the reliability and potential of our method for complex, large-scale applications.

Compared with traditional CNN-based methods such as FC-EF and FC-Siam, which only achieve IoU scores of 64.85% and 66.66% respectively, our model demonstrates a substantial performance gain of more than 20% in IoU. This indicates that the proposed architecture effectively addresses the limitations of conventional convolutional operations in handling complex and heterogeneous remote sensing data. Among more advanced CNN architectures, MSCANet and DAMFANet show relatively competitive performance, with IoU values of 86.38% and 86.11%. However, our method still surpasses them by margins of 1.94% and 2.21% in IoU, and by 1.06% and 1.22% in F1-score, respectively. This confirms the superiority of our multi-scale context modeling strategy in accurately capturing both local and global changes. iT-based approaches such as ELGCNet also provide improvements over early CNN methods, but their IoU (77.03%) and F1-score (87.03%) remain significantly lower than ours, suggesting that pure Transformer structures may suffer from higher complexity and insufficient fine-grained spatial modeling for this task. In contrast, our method achieves both high accuracy and efficiency by combining the strengths of multi-scale feature aggregation with a lightweight ResNet-18 backbone. n addition to the quantitative results, visual inspections (Figure 3) further verify the effectiveness of our approach. Our method produces clearer change maps with fewer false positives and sharper boundaries compared to competing methods. This visual evidence aligns with the higher Precision observed in Table I , indicating that our model is more effective in suppressing pseudo-changes and enhancing true change detection. Furthermore, as shown in the loss curve analysis (Figure 4), the training process exhibits a smooth and stable convergence, demonstrating the reliability and robustness of the optimization process.

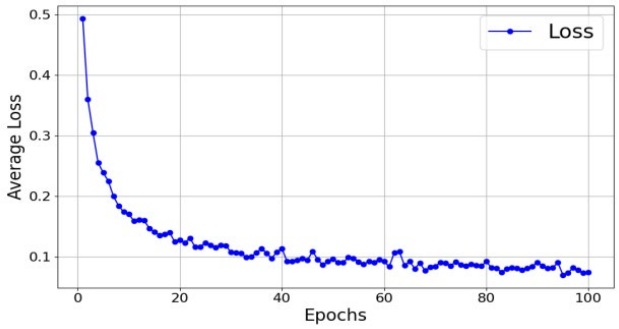


Figure. 4. Loss decline curve.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:45:10 UTC from IEEE Xplore. Restrictions apply.

IV. CONCLUSION

In this work, we proposed a novel GPU-based MSCA module for RSCD, which effectively addresses the challenges of high-resolution image analysis. The MSCA module integrates parallel convolutions with multi-scale dilation rates, image-level pooling, and differential feature processing, providing a powerful tool for capturing contextual information. Unlike Transformer-based models, which suffer from high computational complexity, our approach delivers efficient performance without sacrificing accuracy. Experimental results show existing state-of-the-art RSCD models in terms of key evaluation metrics such as IoU, F1-score, and Precision. Specifically, our method achieved an IoU of 88.32%, F1-score of 93.75%, and Precision of 93.45%, outperforming previous methods with similar backbones by significant margins. dditionally, our loss curve analysis indicates stable convergence during training, highlighting the robustness and reliability of the proposed method. The MSCA module's multi-scale feature extraction capability enables accurate change detection in complex high-resolution scenarios, making it a promising solution for real-world remote sensing applications in urban monitoring, disaster assessment, and ecological management.

In conclusion, our proposed approach offers a computationally efficient and highly accurate solution for Remote Sensing Change Detection (RSCD). By integrating multi-scale convolutional techniques, it effectively addresses the challenges that traditional convolutional models and Transformer-based architectures often encounter. These models typically struggle with balancing computational cost and detection accuracy, especially in the context of high-resolution remote sensing data. Our method leverages the strengths of multi-scale aggregation, enabling it to capture both local and global contextual information without sacrificing efficiency. This ability to handle varying scales of data allows our approach to outperform conventional methods, providing a more robust and scalable solution for real-world RSCD tasks. Overall, the results highlight the potential of combining multi-scale convolution with modern deep learning strategies to significantly improve change detection performance in remote sensing imagery.

REFERENCES

[1] W. Zheng et al., "Feature refinement decomposition and relation preference enhancement for remote sensing change detection," in ICASSP 2025-2025 IEEE international conference on acoustics, speech and signal processing (ICASSP), IEEE, 2025, pp. 1–5.

[2] W. Zheng et al., "Cross-temporal knowledge injection with color distribution normalization for remote sensing change detection," IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2025.

[3] L. Yang, G. Xian, J. M. Klaver, and B. Deal, "Urban land-cover change detection through sub-pixel imperviousness mapping using remotely sensed data," Photogrammetric Engineering & Remote Sensing, vol. 69, no. 9, pp. 1003–1010, 2003.

[4] V. Walter, "Object-based classification of remote sensing data for change detection," ISPRS Journal of photogrammetry and remote sensing, vol. 58, no. 3–4, pp. 225–238, 2004.

[5] H. Zhang, K. Chen, C. Liu, H. Chen, Z. Zou, and Z. Shi, "CDMamba: Remote sensing image change detection with mamba," arXiv preprint arXiv:2406.04207, vol. 4, 2024.

[6] Y. Sun, D. Dai, Q. Zhang, Y. Wang, S. Xu, and C. Lian, "MSCA-net: Multi-scale contextual attention network for skin lesion segmentation," Pattern Recognition, vol. 139, p. 109524, 2023.

[7] M. Noman, M. Fiaz, H. Cholakkal, S. Khan, and F. S. Khan, "ELGC-net: Efficient local-global context aggregation for remote sensing change detection," IEEE Transactions on Geoscience and Remote Sensing, 2024.

[8] Y. Liu, K. Wang, M. Li, Y. Huang, and G. Yang, "Exploring the cross temporal interaction: Feature exchange and enhancement for remote sensing change detection," IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2024.

[9] H. Zhang, H. Chen, C. Zhou, K. Chen, C. Liu, Z. Zou, and Z. Shi, "Bifa: Remote sensing image change detection with bitemporal feature alignment," IEEE Transactions on Geoscience and Remote Sensing, 2024.

[10] Li W, Meng X, Zhao Z, et al. Lot: A transformer-based approach based on channel state information for indoor localization[J]. IEEE Sensors Journal, 23(22): 28205-28219, 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:45:10 UTC from IEEE Xplore. Restrictions apply.